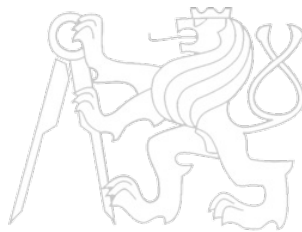


Middleware Architectures 1

Lecture 4: Containers

doc. Ing. Tomáš Vitvar, Ph.D.

tomas@vitvar.com • @TomasVitvar • <https://vitvar.com>



Czech Technical University in Prague

Faculty of Information Technologies • Software and Web Engineering • <https://vitvar.com/lectures>

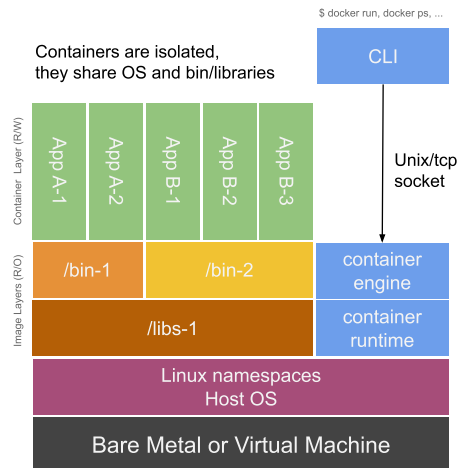
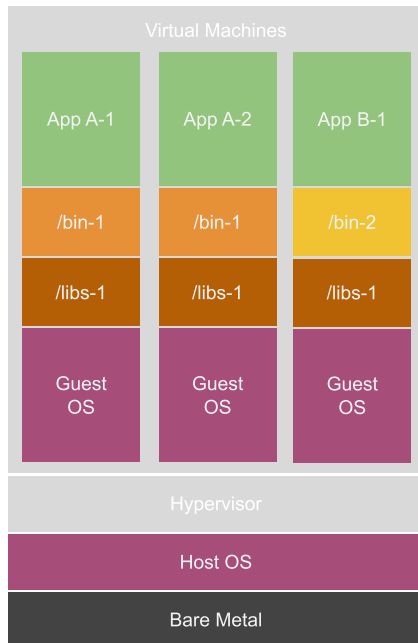


Modified: Mon Oct 27 2025, 06:42:02
Humla v1.0

Overview

- **Overview**
- Linux Namespaces
- Container Image

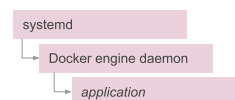
Virtual Machines vs. Containers



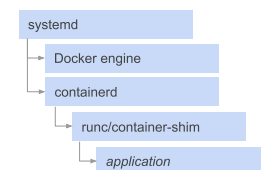
Overview

- **Linux Containers**
 - Introduced in 2008
 - Allow to run a process tree in a isolated system-level "virtualization"
 - Use much less resources and disk space than traditional virtualization
- **Implementations**
 - LXC – default implementation in Linux
 - Docker Containers
 - Builds on Linux namespaces and union file system (OverlayFS)
 - A way to build, commit and share images
 - Build images using a description file called Dockerfile
 - Large number of available base and re-usable images
- **Monolithic design originally**
 - Now several layers
 - container runtime
 - container engine

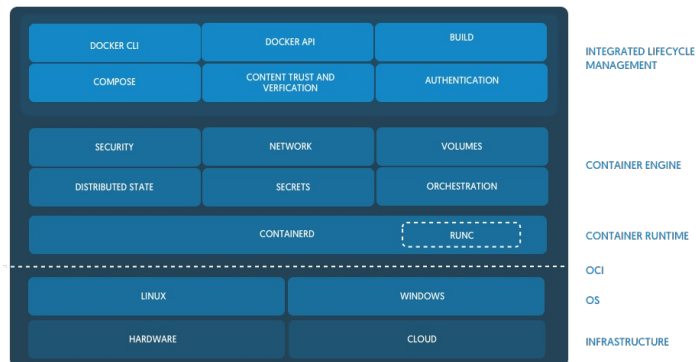
Docker version <1.11.0



Docker version 1.11.0+



Containerd



- Container engine
 - Accepts user inputs (via CLI or API), pulling images from registry, preparing metadata to be passed to container runtime
- Container runtime
 - Abstraction from syscalls or OS specific functionality to run containers on linux, windows, solaris, etc.
 - Uses **runc** and **container-shim**
 - Communicates with kernel to start containerized processes

Terminology

- Image
 - An image contains a union of layered filesystems stacked on top of each other
 - Immutable, it does not have state and it never changes
- Container
 - One or more processes running in one or more isolated namespaces in a filesystem provided by the image
- Container Engine/Runtime
 - The core processes providing container capabilities on a host
- Client
 - An app (e.g. CLI, custom app), communicates with a container engine by its API
- Registry
 - A hosted service containing repository of images
 - A registry provides a registry API to search, pull and push images
 - Docker Hub is the default Docker registry
- Swarm
 - A cluster of one or more docker engines

Overview

- Overview
- **Linux Namespaces**
- Container Image

Linux Namespaces

- Isolation of Linux processes, there are **7 namespaces**
 - *Mount, UTS, IPC, PID, Network, User, Cgroup*
 - *By default, every process is a member of a default namespace of each type*
 - *In case no additional namespace configuration is in place, processes and all their direct children will reside in this exact namespace*
 - *Run **lsns** to check namespaces the process is in*

```
$ lsns
NS      TYPE  NPROCS  PID USER  COMMAND
4026531836 pid    2 30873 oracle -bash
4026531837 user   108 1636 oracle /bin/bash /u01/oracle/scripts/startWebLogicContainer.sl
4026531838 uts     2 30873 oracle -bash
4026531839 ipc     2 30873 oracle -bash
4026531840 mnt     2 30873 oracle -bash
4026531956 net   108 1636 oracle /bin/bash /u01/oracle/scripts/startWebLogicContainer.sl
4026532185 mnt    13 13542 oracle /bin/bash /u01/oracle/scripts/startNM_ohs.sh
4026532192 pid    13 2798 oracle /bin/bash /u01/oracle/scripts/startNM_ohs.sh
...
```

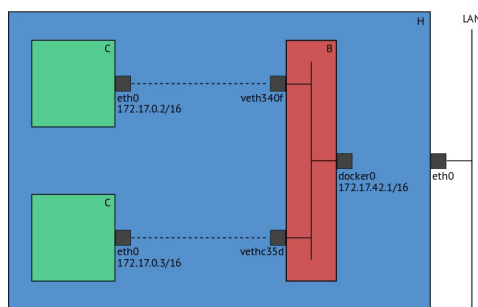
- Flexible configuration, for example:
 - *You can run two apps that only share the network namespace, e.g. **4026531956***
 - *The apps can talk to each other*
 - *Any other app (not in this namespace) won't be able to talk to the apps*

Types: mnt, uts, ipc and pid

- **mnt** namespace
 - Isolates filesystem mount points
 - Restricts the view of the global file hierarchy
 - Each namespace has its own set of mount points
- **uts** namespace
 - The value of the hostname is isolated between different UTS namespaces
- **ipc** namespace
 - Isolates interprocess communication resources
 - message queues, semaphore, and shared memory
- **pid** namespace
 - Isolates PID number space
 - A process ID number space gets isolated
 - Processes can have PIDs starting from the value 1
 - Real PIDs outside of the namespace of the same process is a different number
 - Containers have their own init processes with a PID value of 1

Types: net

- **net** namespace
 - Processes have their own private network stack (interfaces, routing tables, sockets)
 - Communication with external network stack is done by a virtual ethernet bridge



- On the host there is a **userland proxy** or **NAT**
 - NAT is a preferred solution over userland proxy (`/usr/bin/docker-proxy`)
 - Lack of NAT hairpinning may prevent to use NAT
- Use case
 - Multiple services binding to the same port on a single machine, e.g. **tcp/80**
 - A port in the host is mapped to the port exposed by a process in the NS

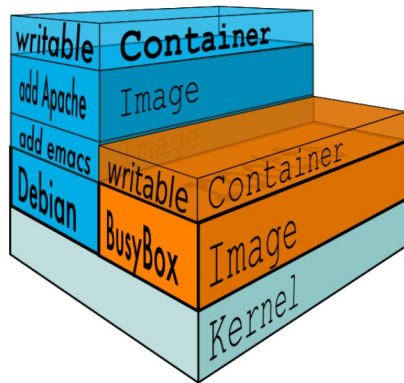
Types: User & Cgroup

- **user** namespace
 - Isolates user and group IDs (UIDs/GIDs) between processes
 - Allows a process to have different privileges inside and outside the namespace
 - Enables **rootless containers** (process is non-root on host, but appears as root inside)
 - Example
 - A process runs as UID 0 (root) inside the container, but maps to a regular UID on the host
- **cgroup** namespace
 - **cgroups** (control groups)
 - Kernel feature to limit and measure process resource usage (CPU, memory, I/O)
 - cgroup namespace
 - Isolates the view of the cgroup hierarchy for each process
 - Prevents a container from seeing/modifying cgroups of the host/other containers
 - Improves security by restricting what resource controls a container can observe
 - **Example:** A container only sees its own CPU/memory usage limits, not the

Overview

- Overview
- Linux Namespaces
- **Container Image**

Container Images



- Containers are made up of R/O layers via a storage driver (OverlayFS, AUFS, etc.)
- Containers are designed to support a single application
- Instances are ephemeral, persistent data is stored in bind mounts or data volume containers.

Image Layering with OverlayFS

- OverlayFS
 - A filesystem service implementing a **union mount** for other file systems.
 - Docker uses **overlay** and **overlay2** storage drivers to build and manage on-disk structures of images and containers.
- Image Layering
 - OverlayFS takes two directories on a single Linux host, layers one on top of the other, and provides a single unified view.
 - Only works for two layers, in multi-layered images hard links are used to reference data shared with lower layers.



Image Layers Example

- Pulling out the image from the registry

```
$ docker pull ubuntu

Using default tag: latest
latest: Pulling from library/ubuntu

5ba4f30e5bea: Pull complete
9d7d19c9dc56: Pull complete
ac6ad7efd0f9: Pull complete
e7491a747824: Pull complete
a3ed95cae02: Pull complete
Digest: sha256:46fb5d001b88ad904c5c732b086b596b92cfb4a4840a3abd0e35d5bb6870585e4
Status: Downloaded newer image for ubuntu:latest
```

- Each image layer has its own directory under `/var/lib/docker/overlay/`.
- This is where the contents of each image layer are stored.

- Directories on the file system

```
$ ls -l /var/lib/docker/overlay/

total 20
drwx----- 3 root root 4096 Jun 20 16:11 38f3ed2eac129654acef11c32670b534670c3a06e483fce313d72e3e
drwx----- 3 root root 4096 Jun 20 16:11 55f1e14c361b90570df46371b20ce6d480c434981cbda5fd68c6ff61
drwx----- 3 root root 4096 Jun 20 16:11 824c8a961a4f5e8fe4f4243dab57c5be798e7fd195f6d88ab06aea92
drwx----- 3 root root 4096 Jun 20 16:11 ad0fe55125ebf599da124da175174a4b8c1878afe6907bf7c7857034
drwx----- 3 root root 4096 Jun 20 16:11 edab9b5e5bf73f2997524eebeac1de4cf9c8b904fa8ad3ec43b35041
```

- The organization of files allows for efficient use of disk space.
- There are **files unique to every layer and hard links to files shared with lower layers**

Dockerfile

- Dockerfile is a script that creates a new image

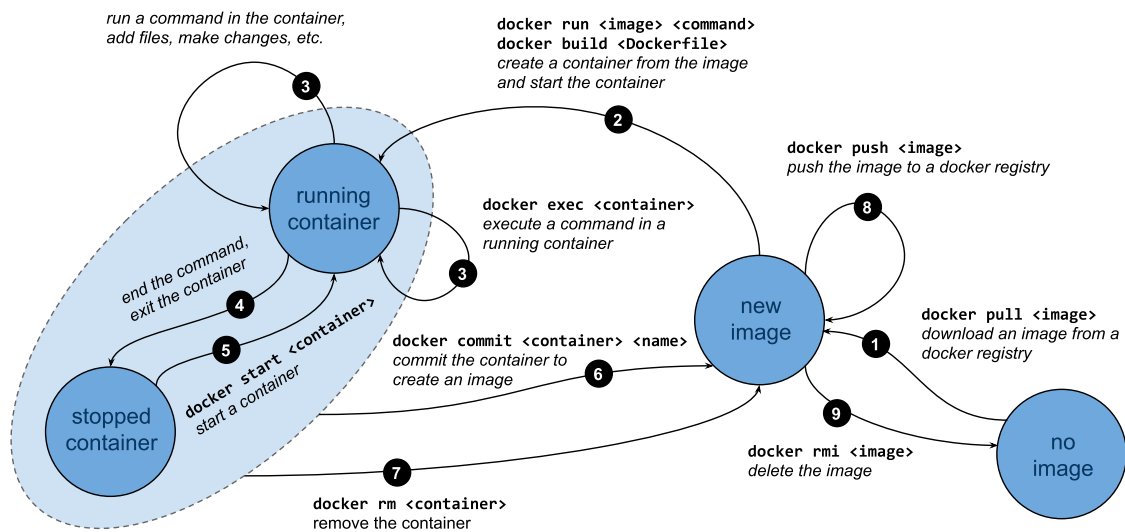
```
# This is a comment
FROM oraclelinux:7
MAINTAINER Tomas Vitvar <tomas@vitvar.com>
RUN yum install -q -y httpd
EXPOSE 80
CMD httpd -X
```

- A line in the Dockerfile will create an intermediary layer

```
$ docker build -t tomvit/httpd:v1 .
Sending build context to Docker daemon 2.048 kB
Step 1 : FROM oraclelinux:7
----> 4c357c6e421e
Step 2 : MAINTAINER Tomas Vitvar <tomas@vitvar.com>
----> Running in 35feebb2ffab
----> 95b35d5d793e
Removing intermediate container 35feebb2ffab
Step 3 : RUN yum install -q -y httpd
----> Running in 3b9aee3c3ef1
----> 888c49141af9
Removing intermediate container 3b9aee3c3ef1
Step 4 : EXPOSE 80
----> Running in 03e1ef9bf875
----> c28545e3580c
Removing intermediate container 03e1ef9bf875
Step 5 : CMD httpd -X
----> Running in 3c1c0273a1ef
```

If processing fails at some step, all preceeding steps will be loaded from the cache on the next run.

Docker Container State Diagram



- 1: There is no image in the local store; you pull an image a remote registry.
- 2: You run a new container on top a specified image.
- 3: You modify the container by adding a library/content in it; you can also run a command in the container from the host.
- 4: You stop a running container.

- 5: You start a stopped container.
- 6: You commit the container and create a new image from it.
- 7: You remove the container.
- 8: You push the image to the remote registry.
- 9: You can remove the image from the local store.